

Codex Prompt: Add Arbitrary-Cycle Carryover as a New Model Option

Toxicity and efficacy models only - trial-design logic must remain unchanged

STRICT SCOPE: Do not change any trial-design code or trial behavior. Only add the new model option, its likelihood/prediction code, and the minimum patient-administration data structure needed to support full cycle history. Existing model options and all allocation/stopping/recycling rules must continue to behave exactly as before.

Task

Add a new arbitrary-cycle geometric carryover model option for both toxicity and efficacy. The model must support patients receiving different numbers of cycles and must preserve each patient's complete treatment-history state.

Do not replace or alter any existing carryover model. Add the multicycle formulation as an additional selectable model option, using names consistent with the current codebase, for example:

```
toxicity_model = "multicycle_additive"
efficacy_model = "multicycle_additive"
```

All existing model options must continue to work exactly as before. The full-history state is used only when the new multicycle option is selected.

1. Arbitrary-Cycle Toxicity Model

For patient i , let $c = 1, \dots, C_i$ denote treatment cycle, where C_i may differ across patients. Let d_{ic} denote the actual dose received by patient i in cycle c , $p_{T,j}$ the regular toxicity probability at dose j , and α_T the toxicity carryover parameter.

Define an uncapped deterministic toxicity state:

$$\begin{aligned} S_{T,i1} &= p_{T,d_{i1}} \\ S_{T,ic} &= p_{T,d_{ic}} + \alpha_T * S_{T,i,c-1}, \quad c \geq 2 \end{aligned}$$

The Bernoulli/event probability is then:

$$q_{T,ic} = \min(1, S_{T,ic})$$

Equivalently, before the final cap:

$$S_{T,ic} = p_{T,d_{ic}} + \alpha_T p_{T,d_{i,c-1}} + \alpha_T^2 p_{T,d_{i,c-2}} + \dots + \alpha_T^{(c-1)} p_{T,d_{i1}}$$

2. Arbitrary-Cycle Efficacy Model

Define the efficacy model analogously, with a distinct efficacy carryover parameter α_E :

$$\begin{aligned} S_{E,i1} &= p_{E,d_{i1}} \\ S_{E,ic} &= p_{E,d_{ic}} + \alpha_E * S_{E,i,c-1}, \quad c \geq 2 \\ q_{E,ic} &= \min(1, S_{E,ic}) \end{aligned}$$

Equivalently:

$$S_{E,ic} = p_{E,d_{ic}} + \alpha_E p_{E,d_{i,c-1}} + \alpha_E^2 p_{E,d_{i,c-2}} + \dots + \alpha_E^{(c-1)} p_{E,d_{i1}}$$

The carryover decay is by number of cycles elapsed, not by numerical dose-level distance. The previous-cycle dose can be the same dose, a higher dose, or a lower dose.

3. Critical State Definition

The recursion must use the uncapped previous state S , not the capped probability q . The state may exceed 1; only the probability used by the likelihood is capped.

```
S_current <- p_current + alpha * S_previous
q_current <- min(1, S_current)
```

Do NOT implement:

```
q_current <- min(1, p_current + alpha * q_previous)
```

That would define a different model because an intermediate cap would truncate historical carryover before the next cycle.

4. Carryover Parameters

Initially implement shared carryover parameters for the new option:

```
alpha_T ~ Beta(a_T, b_T)
alpha_E ~ Beta(a_E, b_E)
```

α_T and α_E are separate parameters and must not be forced to be equal. If the existing framework supports fixed carryover parameters as well as estimated parameters, preserve that option where it can be added without changing trial logic.

5. Minimum Required Administration Data Structure Changes

Make only the data-structure changes required to recover each patient's immediately preceding multicycle state. Keep one row per administration.

Each administration should identify at least:

```
patient_id
admin_id
cycle
dose
previous_admin_id # or equivalent previous_state_index
```

For cycle 1, use an explicit zero-state sentinel. For later cycles, the previous index must point to the immediately preceding administration of the same patient.

Do not assume that the previous row in the global administration table belongs to the same patient, because administrations from different patients can be interleaved in calendar time.

A suitable JAGS-oriented representation is a dummy state at index 1 and a `previous_state_index` for every administration row:

```

S_T[1] <- 0
S_E[1] <- 0

S_T[r + 1] <- p_T[dose[r]] + alpha_T * S_T[previous_state_index[r]]
q_T[r] <- min(1, S_T[r + 1])

S_E[r + 1] <- p_E[dose[r]] + alpha_E * S_E[previous_state_index[r]]
q_E[r] <- min(1, S_E[r + 1])

```

For a patient's first administration, `previous_state_index[r]` must point to the zero-state sentinel. For every later administration, it must point to the state generated by that same patient's previous administration.

6. Complete-Outcome Likelihood

When the multicycle model option is active, use the multicycle probabilities in the ordinary Bernoulli likelihood:

$$Y_{T,ic} \sim \text{Bernoulli}(q_{T,ic})$$

$$Y_{E,ic} \sim \text{Bernoulli}(q_{E,ic})$$

For cycle 1, $q_{T,i1}$ and $q_{E,i1}$ must reduce exactly to the regular dose-specific toxicity and efficacy probabilities.

7. TITE Toxicity Likelihood

Do not change the existing toxicity TITE convention. Only replace the current row probability with $q_{T,ic}$ from the new multicycle model.

$$L_{T,ic} = (w_{T,ic} q_{T,ic})^{Y_{T,ic}} * (1 - w_{T,ic} q_{T,ic})^{(1 - Y_{T,ic})}$$

Therefore the endpoint-specific contributions remain:

- Observed DLT: $q_{T,ic}$, with $w_{T,ic}$ forced to 1.
- Completed non-DLT: $1 - q_{T,ic}$.
- Pending non-DLT: $1 - w_{T,ic} q_{T,ic}$.

Do not replace the pending toxicity contribution with $(1 - q_{T,ic})^{w_{T,ic}}$.

8. Delayed-Outcome Efficacy Likelihood

Do not change the existing delayed-outcome efficacy convention. Use $q_{E,ic}$ from the new multicycle model in the current likelihood.

Let $\Delta_{E,ic} = 1$ when the efficacy outcome for administration (i,c) is ascertained, meaning either a response has occurred or the full efficacy assessment window has completed. Otherwise $\Delta_{E,ic} = 0$.

$$L_{E,ic} = q_{E,ic}^{(\Delta_{E,ic} Y_{E,ic})} * (1 - q_{E,ic})^{[\Delta_{E,ic}(1 - Y_{E,ic}) + (1 - \Delta_{E,ic}) w_{E,ic}]}$$

Therefore the endpoint-specific contributions remain:

- Observed efficacy response: $q_{E,ic}$.

- Completed nonresponse: $1 - q_{E,ic}$.
- Pending nonresponse: $(1 - q_{E,ic})^{w_{E,ic}}$.

Preserve the existing zeros-trick or custom-likelihood implementation if that is how pending efficacy observations are currently handled.

9. Simulation Data-Generating Mechanism

When the new multicycle model option is selected in simulation, generate outcomes from the same full-history recursive state. Maintain a true uncapped state for each patient separately.

```

true_S_T_current <- p_true[current_dose] + alpha_T_true * true_S_T_previous
true_q_T_current <- min(1, true_S_T_current)

true_S_E_current <- e_true[current_dose] + alpha_E_true * true_S_E_previous
true_q_E_current <- min(1, true_S_E_current)

```

For cycle 1, the previous true state is zero. Do not generate later cycles using only the immediately previous dose probability; the full uncapped state must be propagated.

10. Prediction for a Proposed Next Cycle

When the existing code requests model-specific toxicity or efficacy probabilities for a proposed recycled/next-cycle administration at dose k , use the patient's current posterior multicycle state.

$$\begin{aligned}
 S_{T,i,c+1}(k) &= p_{T,k} + \alpha_T S_{T,ic} \\
 q_{T,i,c+1}(k) &= \min(1, S_{T,i,c+1}(k)) \\
 S_{E,i,c+1}(k) &= p_{E,k} + \alpha_E S_{E,ic} \\
 q_{E,i,c+1}(k) &= \min(1, S_{E,i,c+1}(k))
 \end{aligned}$$

Use these model-specific probabilities wherever the existing model interface requires a toxicity or efficacy probability for the proposed next administration. Do not alter the recycling decision criterion itself.

11. Non-Negotiable Scope Restriction

Do not refactor, reinterpret, or modify the trial design. In particular, do not change any allocation, admissibility, no-skipping, Stage I/Stage II, futility, overtotoxicity, OBD, cohort, event-clock, n_{eval} , recycling eligibility, enrollment-priority, or stopping logic.

The goal is only to add a new multicycle toxicity model option and a new multicycle efficacy model option, connect them through the existing model-selection interface, and make the minimum patient-history/data-structure changes required for those models.

If any existing trial-design function would need to be changed for reasons beyond passing the new model-specific probabilities/state information through the current interface, flag that explicitly rather than modifying the design logic.

12. Implementation Summary Requested from Codex

After implementation, report only:

Arbitrary-Cycle Carryover Model

- the new model-option names;
- the files modified;
- the administration/patient-history fields added or changed;
- the exact toxicity and efficacy state recursion implemented;
- how the previous-state index is constructed;
- how the new model enters complete-outcome, TITE toxicity, and delayed-outcome efficacy likelihoods;
- how the next-cycle patient-specific probabilities are calculated; and
- any place where the model could not be added without touching trial-design logic.